

Article

Sustainable Digital Transformation in Music Education: An Analysis of Teacher Competencies in the Light of TPACK and International Frameworks [†]

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Abstract

The education systems, financial circumstances, and societal structures of our century expect educators to possess the most important characteristic: the ability to guide students who are highly digitally competent and keep themselves up to date. The “Sustainable Development Goals (SDG 4)” emphasized by the According to the United Nations highlight the necessity of continuously updating teacher competencies for quality and inclusive education. Establishing music teachers’ “digital competencies” on a sustainable basis depends on combining technical skills with a pedagogical vision. Therefore, thoroughly examining music teachers’ digital competencies in light of international standards and the TPAC model is critical to ensuring the sustainability of digital transformation at both the institutional and individual levels. This study, which examines digital literacy as an important part of sustainable education in music education, has examined the digital skills of music teachers in Turkey within the scope of international digital literacy frameworks and the TPAC approach. Digital skills have been related to the status of teachers’ professional practices, teaching-learning processes, assessment approaches, and the support of students’ digital literacy. The research concluded that music teachers’ digital competency levels are at the “explorer” level, meaning they are individuals who are aware of digital technologies and conduct research to develop themselves in this area.

Keywords: digital competence; music teachers; teacher training; educational technologies; TPAC



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1. Introduction

Digitalization has emerged as a defining force in the social, economic, and cultural domains of the twenty-first century, forcing educational systems to abandon conventional methods of imparting knowledge [1,2]. One of the pillars of the “UN 2030 Agenda for Sustainable Development”, the “Sustainable Development Goals (SDG 4)” is to guarantee

accessible, equitable, and high-quality education for everyone and to encourage opportunities for lifelong learning [3,4]. The most important prerequisite and main factor in accomplishing these worldwide goals is acknowledged to be education [4,5]. However, the pace of the digital revolution has not yet kept up with the evolution of educational institutions and school teaching–learning procedures [6].

Teachers are seen as the primary players, strategists, and designers who organize and direct learning environments during the digital transformation process [6–8]. Teachers are expected by modern educational institutions to take on roles as organizers of the learning process, data analyzers, and pedagogical designers in addition to being knowledge transmitters [7,8]. Music education is one of the most exciting areas of application for this transition because of its special nature, where creative creation is combined with digital innovation [9]. Even while music educators are highly motivated to use technology, there are substantial institutional obstacles in the way of turning their abilities into a long-term pedagogical model [9].

Studies that have already been published in the literature [10–14] show that teachers' levels of digital competence typically stay at moderate or low levels and that they struggle, especially when it comes to creating digital content, solving problems, and integrating technology into the pedagogical process. The profile of teachers' digital literacy is primarily tool- and information-focused, falling short of the depth needed to produce original knowledge [13]. Additionally, it is emphasized that using technology in the classroom does not directly improve learning outcomes; rather, the true difference comes from using technology with “pedagogical depth” [6,14,15].

Teachers must integrate their technological expertise with a pedagogical vision for the digital transition to be sustainable. In this sense, the “TPACK” paradigm offers a basic theoretical framework that explains how content knowledge (K), pedagogy (P), and technology (T) should function in concert [15,16]. Additionally, strategic guidelines that specify “how and to what standard” these competences should be used are provided by international standards as “DigCompEdu” and the “UNESCO ICT” [17] teacher competency framework [18–21]. Teachers' TPACK levels and performance self-assessments are strongly correlated, according to the research [21–23]; this correlation becomes even more significant when considering sustainable education trends and digital competency levels.

Teachers must therefore continuously “update” their digital literacy due to the quick obsolescence of technological equipment and the unchecked growth of digital resources, which puts the sustainability of educational quality at risk [9,23]. In order to ensure that digital transformation is structured not just as the acquisition of technical skills but as an ethical, inclusive, and sustainable pedagogical approach, it is crucial to analyze music teachers' digital competencies in light of both the TPACK framework and international standards while taking into account the particular structure of music education [15,18,24,25].

1.1. Theoretical/Conceptual Framework

1.1.1. The Framework for “Technological Pedagogical Content Knowledge (TPACK)”

“The Technological Pedagogical Content Knowledge (TPACK)” model, created by Mishra and Koehler [26], is the most basic theoretical framework for comprehending teachers' roles in digital transformation [6,8,9]. The dynamic relationship between technology (T), pedagogy (P), and content knowledge (K) is explained by the seven components of TPACK [9,18]. In the literature, TPACK is defined as a conceptual foundation that evaluates not only the use of technological tools but also how and why these tools should be applied in a pedagogically sound and content-appropriate manner [26]. The self-efficacy theory, developed by Bandura, is recognized as a critical psychological factor that determines

teachers' motivation to incorporate digital tools into their teaching processes and their resilience in the face of technological challenges [27].

This paradigm states that a music teacher must know how to integrate technical tools (TK) with musical content (CK) and how to use pedagogical methods (PK) to increase student comprehension [12,18]. TPACK is presented as a “new form of application literacy” that transcends conventional conceptions of literacy in the literature [11,12]. Research shows that teachers with high TPACK competence are better able to design student-centered and innovative teaching processes that align with educational reforms [25].

1.1.2. Global Frameworks for Digital Competence: UNESCO and DigCompEdu

The conceptual boundaries of this study are defined by two important models created to standardize educators' digital competencies. The European Digital Competence Framework for Educators, or DigCompEdu, requires educators to have 22 competencies in six areas: professional commitment, digital resources, teaching and learning, evaluation, learner empowerment, and supporting students' digital competencies (Figure 1). In this context, digital literacy is viewed as the foundation of inclusive education. Teachers' digital literacy enables them to adapt and personalize learning processes according to individual student needs, thereby allowing them to view diversity as a strength [28,29]. Research shows that a high level of digital literacy is essential for the sustainable development of pre-service teachers' professional competencies [30]. According to this approach, the teacher is a “strategist” who creates digital learning procedures rather than just a user [31].

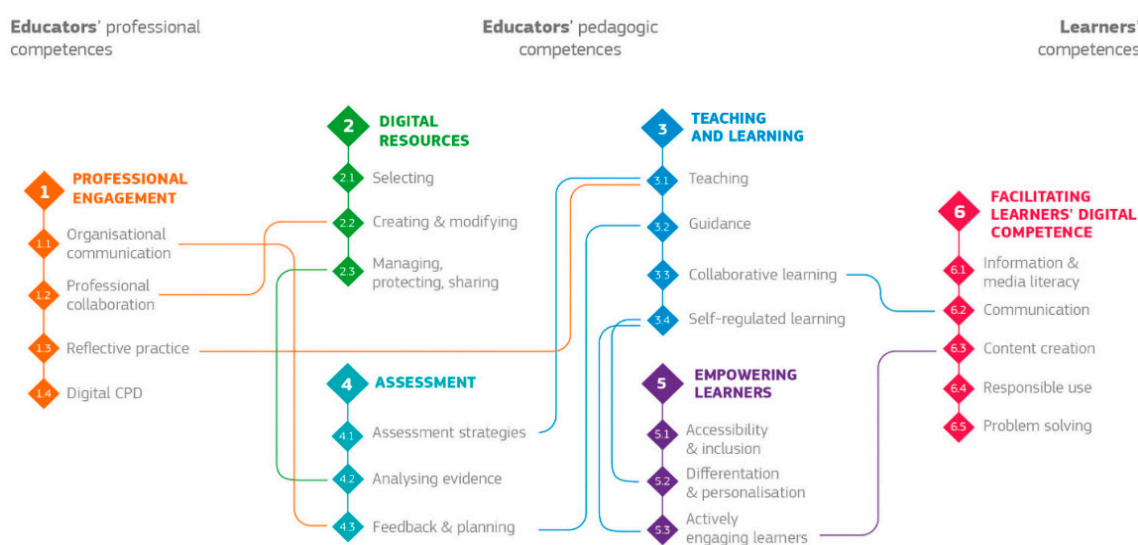


Figure 1. The European Framework for the Digital Competence of Educators (DigCompEdu) details 22 educator-specific digital competencies, organized in six areas [31].

This framework evaluates educators across six distinct levels, ranging from “new-comer” (A1) to “pioneer” (C2), and positions them as strategists for the complex learning environments required by the digital age [25]. These levels and their characteristics [3] are listed below:

- A1 and A2 (Beginner Level): Teachers at this level are aware of the educational potential of digital tools but have only integrated technology into their lessons to a limited extent.
- B1 and B2 (Intermediate Level): This covers educators who incorporate technology into their teaching processes in a meaningful and purposeful way, and who systematically enhance their digital strategies.

- C1 and C2 (Advanced Level): This is the highest level, comprising educators who share their expertise with colleagues, devise complex digital strategies, and lead pedagogical innovation to develop new models.

1.2. “UNESCO ICT-CFT (ICT” Competency Framework for Teachers)

It takes a three-stage approach to teacher development: knowledge generation, knowledge deepening, and knowledge acquisition. The highest level for music educators, “knowledge creation,” stands for creative leadership that empowers students to create unique musical endeavors. The “DigCompEdu” and UNESCO [17] frameworks specify “how and to what standard” these competencies should be applied, while the TPACK model specifies “what they need to possess.” By combining these two frameworks, music educators can develop a level of professional competence that is long-lasting and more resistant to technological advancements.

1.3. The Principal Effects of Digitalization on the Roles of Teachers

By encouraging a shift away from conventional methods in educational systems, digitalization is radically changing the roles and methods of instruction of instructors [1,2]. This change involves reorganizing educational goals, learning environments, and pedagogical approaches in addition to using technology as a tool [1,2]. In the digital age, the job of the teacher has changed from being a “transmitter of knowledge” in the traditional education model to a multifaceted identity [3];

- Guide and Facilitator (Guide on the Side): The instructor is no longer the “sage on the stage,” but rather a guide who helps students access information and solve difficulties [3,4]. The social constructivist approach proposed by Vygotsky emphasizes that learning is a process of social interaction and argues that teachers should guide students in discovering knowledge by providing scaffolding within the “zone of proximal development” [3,6]. The modern teacher is defined as both a data analyst who monitors student performance based on learning analytics derived from digital platforms and a strategist who designs dynamic learning environments [2].
- Educational Designer and Strategist: Teachers are positioned as designers and strategists who modify “digital tools and resources” to create dynamic learning environments and satisfy students’ requirements [3,5,6].
- Data Analyst: Today’s educators take on the job of an analyst, monitoring student performance and creating interventions based on data from digital platforms and learning analytics [3,6].
- Mentor and Facilitator: They serve as mentors, assisting students in acquiring skills related to critical thinking, creativity, and lifelong learning [6].

1.4. How Educational Technology Can Help Achieve the “Sustainable Development Goals”

In order to achieve the “2030 Sustainable Development Goals (SDGs)” of the United Nations, especially SDG 4 on “Quality Education,” educational technologies are positioned as a transformative force and a crucial component [1,2]. SDG 4 aims to ensure inclusive and equitable quality education that promotes lifelong learning for all [3,16,30–32].

A comprehensive reorganization of teaching and learning paradigms is necessary for the process of sustained digital transformation in music education, rather than just bringing technology into the classroom as a tangible tool [1,2]. Teachers are no longer the only source of knowledge in contemporary educational systems; instead, they now play the role of strategists and mentors who create and oversee digital learning procedures and mold learning environments to meet the unique needs of each student [3,4]. The fifth area

of the DigCompEdu paradigm, “empowering learners,” is a pivotal element in converting digital competencies into pedagogical success and sustainability.

The domain of “empowering learners” emphasizes fostering inclusion in education, personalizing learning experiences, and promoting students’ active engagement in lessons using digital tools [4,6]. From the viewpoint of music educators, this competency involves employing assistive technologies to enhance access to musical resources for students with disabilities or special educational needs, in addition to creating digital content (such as e-sheet music and interactive solfège software) customized to each student’s unique learning pace [6]. Research [7,8] indicates that student-centered methodologies not only improve academic performance but also bolster students’ self-efficacy beliefs and engagement with the subject matter. The sustainability of digital transformation relies not solely on teachers’ technical skills (TS) but is fundamentally connected to their “TPACK”, which encompasses the integration of these skills with pedagogical and subject-specific knowledge [9,10].

In accordance with the “United Nations Sustainable Development Goals (SDG 4—Quality Education)”, it is imperative for educators to perpetually enhance their “digital competencies” and use these skills with an ethical, responsible, and inclusive perspective [2,3,14]. Empirical research suggests that teachers with a high level of digital competence and a dedication to sustainable education perform better in performance self-assessments and may better arrange their learning environments [8,15].

1.5. Research Gap

A study of the available literature suggests that the majority of digital transformation research focuses on global trends, national plans, or organizational practices in isolation [1]. Although global digital competence models have been thoroughly described on a theoretical level, research into the systematic integration of these models with national education reforms and processes is scarce [2]. A major proportion of existing digital competency frameworks focus on the use of technology in the classroom and in face-to-face situations [3].

Another notable gap in the literature on digital competence is the move from broad definitions to “subject-specific” competencies [3]. Despite the fact that music education is a subject that integrates artistic performance, creativity, and technical abilities, a thorough examination of music teachers’ competencies in light of both the TPACK framework and current international standards is uncommon. While existing studies assess instructors’ technical tool-use skills (TK), there is a scarcity of evidence-based data on how technology is linked with musical pedagogy and the sustainability mission at a “pedagogical depth” [3,4]. Furthermore, the literature generally limits the link between digitalization and the “Sustainable Development Goals (SDGs)” to the theoretical level; there is a noticeable lack of empirical models demonstrating how digital skills support long-term resilience and sustainable transformation of the education system [4,5].

According to recent study, music teachers are highly motivated to use technology, but they confront institutional difficulties in turning these talents into a viable educational model [33]. The quick obsolescence of technology equipment creates a perpetual demand for teachers to “upgrade” their digital competencies, putting educational quality at risk. As a result, it is deemed necessary to conduct a thorough examination of music teachers’ digital competences in light of international standards and the TPACK model, in order to assure the long-term viability of digital transformation at both institutional and individual levels.

This study aims to fill a critical gap in the literature by assessing music teachers’ digital competencies using a multi-layered analysis (TPACK and international frameworks), demonstrating how digitalization can be structured as a sustainable and inclusive pedagogical revolution rather than a technical update.

The purpose of this research is to examine music teachers' "digital competencies" in light of the Technological Pedagogical Content Knowledge (TPACK) model and international competence frameworks such as DigCompEdu, thereby identifying the opportunities and constraints of long-term digital transformation in music education. The study intends to illustrate how teachers' technical skills interact with the special pedagogical requirements of music instruction. In this regard, the study links digital abilities to the state of instructors' professional practices, teaching and learning processes, assessment methodologies, and support for students' digital competencies. As a result, the research problem and subproblems are presented below.

RQ1. What are the digital competence levels of music teachers?

RQ2. Do the digital competence levels of music teachers differ in the sub-dimension of "use of digital skills in the profession"?

- According to the status of possessing internet and technological materials?
- According to the level of "TPACK"?

RQ3. Do the digital competence levels of music teachers differ in the sub-dimension of "digital resources"?

- According to the status of possessing internet and technological materials?
- According to the level of "TPACK"?

RQ4. Do the digital competence levels of music teachers differ in the sub-dimension of "teaching and learning"?

- According to the status of possessing internet and technological materials?
- According to the level of "TPACK"?

RQ5. Do the digital competence levels of music teachers differ in the sub-dimension of "assessment"?

- According to the status of possessing internet and technological materials?
- According to the level of "TPACK"?

RQ6. Do the digital competence levels of music teachers differ in the sub-dimension of "empowering learners"?

- According to the status of possessing internet and technological materials?
- According to the level of "TPACK"?

RQ7. Do the digital competence levels of music teachers differ in the sub-dimension of "facilitating students' digital competence"?

- According to the status of possessing internet and technological materials?
- According to the level of "TPACK"?

2. Materials and Methods

2.1. Research Design

The research is a "descriptive study" aimed at examining the digital competences of music teachers. In our study, which was prepared using the quantitative research method, the general survey model was employed. "General survey models are survey arrangements conducted on the whole population or on a group, sample, or subset taken from it, in a population consisting of a large number of elements, in order to reach a general judgment about the population" [34] (p. 111).

2.2. Population and Sample of the Research

The population of the research consists of music teachers working in different types of schools affiliated with the Ministry of National Education in the cities of Mersin, Bursa,

Niğde, Van, Şırnak, Rize, and Muğla, selected to represent the seven geographical regions of Türkiye during the 2023–2024 academic year. The sample consists of 565 music teachers who were easily accessible. The convenience sampling method is a low-cost research method through which the maximum output can easily be obtained from the topics that need to be examined in the sample [35]. Reaching the entire population requires excessive time and cost. Therefore, in order to conduct the research process in a sound manner, the convenience sampling method was used.

In the Table 1, the demographic information of the 565 music teachers who participated in the research is presented.

Table 1. Demographic information of the teachers participating in the research.

Demographic Information		<i>f</i>	%
Gender	Female	301	53.3
	Male	264	46.7
	Total	565	100
Age	20–30 years	77	13.6
	30–40 years	272	48.1
	40–50 years	188	33.3
	50 years and above	28	5.0
	Total	565	100
Educational Level	Bachelor’s Degree	532	94.2
	Master’s Degree	29	5.1
	Doctoral Degree	4	0.7
	Total	565	100
Total Years of Professional Experience	Less than 5 years	61	10.8
	5–10 years	168	29.7
	10–15 years	126	22.3
	15–20 years	98	17.3
	20 years and above	112	19.8
	Total	565	100
Type of School Employed	Public Secondary School	413	73.1
	Public Anatolian High School	44	7.8
	Public Social Sciences High School	3	0.5
	Public Science High School	4	0.7
	Public Vocational and Technical Anatolian High School	22	3.9
	Public Science and Art Center	5	0.9
	Public Imam Hatip Anatolian High School	24	4.2
	Public Imam Hatip Anatolian Secondary School	29	5.1
	Public Fine Arts High School	21	3.7
	Total	565	100

Table 1. Cont.

Demographic Information		<i>f</i>	%
Status of Possessing Internet and Technological Materials	Internet only	34	6.0
	Technological material/materials only	9	1.6
	I possess both	522	92.4
	Total	565	100
TPACK Level	Low	133	23.5
	Moderate	273	48.3
	High	159	28.1
	Total	565	100
Participation in Training Programs	Participated	285	50.4
	Didn't participate	280	49.6
	Total	565	100

2.3. Data Collection Instruments

2.3.1. Digital Competence Scale for Educators

The “digital competence scale for educators” was developed in 2017 by Redecker [31] on behalf of the European Union to determine the levels of digital competences of educational staff and to ensure their development. Toker et al. [36] adapted this scale into Turkish in order to support the development of teachers in our country. In this scale, the competences that educational personnel are expected to possess are addressed under 6 headings and 22 items. “As a result of the implementation of the scale, participants are evaluated under six categories: newcomer, explorer, integrator, expert, leader, and pioneer” [36] (p. 307).

Toker et al. [36] used confirmatory factor analyses to evaluate the scale’s concept validity and factor structure. According to their report, the first test yielded a χ^2 value of 1729.284 with $df = 194$ and fit indices of RMSEA = 0.048, RMR = 0.041, CFI = 0.96, and GFI = 0.95. The χ^2 value was measured as 1387.40 with $df = 190$ in the second test after the changes, and the fit indices were found to be RMSEA = 0.043, RMR = 0.038, CFI = 0.98, and GFI = 0.96. These findings led to the conclusion that the data set suited the model quite well. The correlation coefficient between the scores from two assessments conducted at different times was determined by Toker et al. [36] as 0.78 after a test–They found that the test reliability level was high after doing a test–retest procedure for scale reliability. Additionally, they discovered that the scale had a high degree of internal consistency, with a Cronbach’s Alpha of 0.94, and that the item–total score correlations, as shown by item analysis, varied from $r(j(x))$ 0.488 to 0.710. These findings suggest that the items of the “digital competence scale for educators” have adequate discriminant validity. Within the scope of this study, the reliability of the scale was found to be high ($\alpha = 0.93$).

2.3.2. Personal Information Form

The “personal information form” was prepared in order to interpret the data obtained from the scale through the demographic information of the participants within their own consent form. The form includes information regarding the purpose and implementation of the study. It consists of questions about “gender”, “age”, “educational status”, “years of professional experience”, “type of institution” where they work, the status of possessing technological materials, TPACK levels, and the training they have received for their professional development.

2.4. Data Collection Process

In the data collection process of the research, first, the necessary ethics committee approval was obtained from the Mersin University Social and Humanities Ethics Committee.

Since the research covers teachers selected to represent the seven geographical regions of Türkiye, the necessary permissions were obtained from the Ministry of National Education and from the researchers who adapted the scale used in the study into Turkish. In order to prevent data loss during the data collection process, the “digital competence scale for educators” and the “personal information form” to be used in the study were delivered to and collected from the participants via an online platform (Google Forms link).

2.5. Data Analysis

In this research, arithmetic mean, percentage, frequency analysis, standard deviation, and non-parametric tests were conducted in the data analysis stage. The data obtained in the research were subjected to normality analysis. As a result of the analysis, since the data did not show normal distribution, the “Kruskal–Wallis H” test was used in the analysis of the participants’ status of possessing internet and technological materials and the TPACK variables. The responses given by the participants to the scale were matched according to the scoring criteria determined in the scale, and the digital competence level of the sample was determined. The digital proficiency levels of educators are categorized according to the six-stage hierarchical framework presented in Table 2, based on the DigCompEdu model [31].

Table 2. The digital proficiency levels of educators based on the “DigCompEdu” model.

Level	Level Description	Score Range	Key Characteristics
A1	“Newcomer”	0–19	Limited awareness of digital tools and a significant need for guidance.
A2	“Explorer”	20–33	Limited digital usage and a test period for basic application tools.
B1	“Integrator”	34–49	Integrating technology into teaching methods and making the switch to regular use.
B2	“Expert”	50–65	Implementing digital strategies in a critical and diversified manner.
C1	“Leader”	66–80	Guidance and strategic leadership for colleagues through knowledge sharing.
C2	“Pioneer”	81–100	Developing innovative pedagogical approaches and organizational transformation.

3. Results

1. Related to the “Digital Competence” Levels of Music Teachers

According to the data presented in Table 3, it is observed that the music teachers constituting the sample are at the “explorer” level with a mean score of $\bar{X} = 33.65$ within the framework of “digital competence for educators (DigCompEdu).”

Table 3. The “digital competence” level of the research sample within the framework of the “digital competence scale for educators (DigCompEdu).”

	<i>f</i>	$\bar{X}$
Digital Competence Level of the Sample within the DigCompEdu Framework	565	33.65

2. Findings Related to the Sub-Dimension of “Use of Digital Skills in the Profession” of Music Teachers’ Digital Competence Levels

3.1. Findings Related to the Variable of Status of Possessing Internet and Technological Materials in the Sub-Dimension of “Use of Digital Skills in the Profession”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “use of digital skills in the profession” show a statistically significant difference according to the variable of status of possessing internet and technological materials, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

When Table 4 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “use of digital skills in the profession” show a statistically significant difference in terms of the variable of the status of possessing internet and technological materials ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “use of digital skills in the profession”, music teachers who possess only the internet have statistically higher digital competences compared to other music teachers.

Table 4. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “use of digital skills in the profession” in terms of the variable of the status of possessing internet and technological materials.

Status of Possessing Internet and Technological Materials	<i>n</i>	so	sd	χ^2	<i>p</i>
Internet Only	34	293.87			
Technological Material Only	9	35.50	2	21,410	0.001 *
Both	522	286.56			
Total	565				

* Significant *p*-value at 0.05 level.

3.2. Findings Related to the Variable of “Technological Pedagogical Content Knowledge (TPACK)” Level in the Sub-Dimension of “Use of Digital Skills in the Profession”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “use of digital skills in the profession” show a statistically significant difference according to the variable of “TPACK” level, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

In Table 5, it is seen that the digital competence levels of music teachers in the sub-dimension of “use of digital skills in the profession” show a statistically significant difference in terms of the variable of “TPACK” level ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “use of digital skills in the profession,” music teachers with a high level of “TPACK” have statistically higher “digital competences” compared to others.

Table 5. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “use of digital skills in the profession” in terms of the variable of “Technological Pedagogical Content Knowledge (TPACK)” level.

Technological Pedagogical Content Knowledge (TPACK) Level	<i>n</i>	so	sd	χ^2	<i>p</i>
Low	133	161.03			
Medium	273	261.45	2	197,323	0.001 *
High	159	422.03			
Total	565				

* Significant *p*-value at 0.05 level.

According to the results of the “Kruskal–Wallis H” test, a statistically significant difference was found between the participants’ scores based on their TPACK levels ($\chi^2(2) = 197,323$;

$p = 0.001$). In order to determine the source of this difference, Dunn's post hoc test with Bonferroni correction was conducted. The results indicated that all levels—low, medium, and high—differed significantly from one another ($p < 0.0166$). When the mean ranks were examined, it was observed that the high-TPACK-level group ($MR = 422.03$) had significantly higher scores than both the medium ($MR = 261.45$) and low ($MR = 161.03$) groups. Similarly, the scores of the medium-level group were significantly higher than those of the low-level group. This finding demonstrates that as the TPACK level increases, the related scores also show a consistent and significant increase.

3.3. Findings Related to the Variable of Status of Possessing Internet and Technological Materials in the Sub-Dimension of "Digital Resources"

In order to determine whether the digital competence levels of music teachers in the sub-dimension of "digital resources" show a statistically significant difference according to the variable of status of possessing internet and technological materials, the Kruskal–Wallis H test, one of the non-parametric tests, was applied.

According to Table 6, it is seen that the digital competence levels of music teachers in the sub-dimension of "digital resources" show a statistically significant difference in terms of the variable of status of possessing internet and technological materials ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of "digital resources," music teachers who possess only the internet have statistically higher digital competences compared to other music teachers.

Table 6. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of "digital resources" in terms of the variable of the status of possessing internet and technological materials.

Internet and Technological Material Ownership Status	<i>n</i>	so	sd	χ^2	<i>p</i>
Only Internet	34	350.56			
Only Technological Material	9	64.56	2	22,143	0.001 *
Both	522	282.37			
Total	565				

* Significant *p*-value at 0.05 level.

3.4. Findings Related to the Variable of "Technological Pedagogical Content Knowledge (TPACK)" Level in the Sub-Dimension of "Digital Resources"

In order to determine whether the digital competence levels of music teachers in the sub-dimension of "digital resources" show a statistically significant difference according to the variable of "TPACK" level, the "Kruskal–Wallis H" test, one of the non-parametric tests, was applied.

In Table 7, it is seen that the digital competence levels of music teachers in the sub-dimension of "digital resources" show a statistically significant difference in terms of the variable of "TPACK" level ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of "digital resources," music teachers with a high level of "TPACK" have statistically higher digital competences compared to other music teachers.

The "Kruskal–Wallis H" test was conducted to determine if participants' scores significantly differed based on their TPACK levels. The results revealed a statistically significant difference across the groups ($\chi^2(2) = 158,836$; $p = 0.001$). To identify the specific source of these differences, "Dunn's post hoc test" with "Bonferroni" correction was applied. The post hoc analysis indicated that all three groups—low, medium, and high—differed significantly from each other ($p < 0.0166$). Upon examining the mean ranks, it was found that the "high" TPACK level group ($MR = 395.81$) had significantly higher scores than both

the “medium” (MR = 279.50) and “low” (MR = 155.32) groups. Furthermore, the medium-level group scored significantly higher than the low-level group. These findings suggest a positive and significant relationship between TPACK levels and the measured scores.

Table 7. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “digital resources” in terms of the variable of “Technological Pedagogical Content Knowledge (TPACK)” level.

(TPACK) Level	<i>n</i>	so	sd	χ^2	<i>p</i>	Post Hoc ^b
Low	133	155.32				
Medium	273	279.50				
High	159	395.81	2	158,836	0.001 *	2–1
Total	565					3–1/3–2

^b Dunn’s test with Bonferroni correction. * Significant *p*-value at 0.05 level.

4. Findings Related to the Sub-Dimension of “Teaching and Learning” of Music Teachers’ “Digital Competence” Levels

4.1. Findings Related to the Variable of Status of Possessing Internet and Technological Materials in the Sub-Dimension of “Teaching and Learning”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “teaching and learning” show a statistically significant difference according to the variable of status of possessing internet and technological materials, the “Kruskal–Wallis H” test, one of the “non-parametric” tests, was applied.

When Table 8 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “teaching and learning” show a statistically significant difference in terms of the variable of status of possessing internet and technological materials ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “teaching and learning,” music teachers who only have internet access have statistically higher “digital competences” compared to other music teachers.

Table 8. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “teaching and learning” in terms of the variable of the status of possessing internet and technological materials.

Internet and Technological Material Ownership Status	<i>n</i>	so	sd	χ^2	<i>p</i>
Only Internet	34	294.46			
Only Technological Material	9	119.44	2	9360	0.009 *
Both	522	285.07			
Total	565				

* Significant *p*-value at 0.05 level.

4.2. Findings Related to the Variable of “Technological Pedagogical Content Knowledge (TPACK)” Level in the Sub-Dimension of “Teaching and Learning”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “teaching and learning” show a statistically significant difference according to the variable of “TPACK” level, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

In Table 9, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “teaching and learning” show a statistically significant difference in terms of the variable of “TPACK” level ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “teaching and learning,” music teachers with a high level of “TPACK” have statistically higher “digital competences” compared to other music teachers.

Table 9. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “teaching and learning” in terms of the variable of “Technological Pedagogical Content Knowledge (TPACK)” level.

(TPACK) Level	<i>n</i>	so	sd	χ^2	<i>p</i>	Post Hoc ^b
Low	133	218.74	2	84,865	0.001 *	3–1/3–2
Medium	273	257.44				
High	159	380.65				
Total	565					

^b Dunn’s test with Bonferroni correction. * Significant *p*-value at 0.05 level.

According to the results of “Dunn’s” test with “Bonferroni” correction, which was conducted to identify the source of the significant difference found in the Kruskal–Wallis H” test ($\chi^2(2) = 84,865$; $p = 0.001$), it was determined that the “teaching and learning” scores of the “high” TPACK level group were significantly higher than both the medium and low-level groups ($p < 0.001$). However, no statistically significant difference was found between the low- and medium-level groups ($p > 0.0166$).

5. Findings Related to the Sub-Dimension of “Assessment” of Music Teachers’ “Digital Competence” Levels

5.1. Findings Related to the Variable of Status of Possessing Internet and Technological Materials in the Sub-Dimension of “Assessment”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “assessment” show a statistically significant difference according to the variable of status of possessing internet and technological materials, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

When Table 10 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “assessment” do not show a statistically significant difference in terms of the variable of status of possessing internet and technological materials ($p > 0.05$).

Table 10. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “assessment” in terms of the variable of the status of possessing internet and technological materials.

Internet and Technological Material Ownership Status	<i>n</i>	so	sd	χ^2	<i>p</i>
Only Internet	34	291.68	2	1766	0.414
Only Technological Material	9	215.11			
Both	522	283.61			
Total	565				

5.2. Findings Related to the Variable of “Technological Pedagogical Content Knowledge (TPACK)” Level in the Sub-Dimension of “Assessment”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “assessment” show a statistically significant difference according to the variable of “TPACK” level, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

When Table 11 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “assessment” show a statistically significant difference in terms of the variable of “TPACK” level ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “assessment,” music teachers with a high level of “TPACK” have statistically higher “digital competences” compared to other music teachers.

Table 11. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “assessment” in terms of the variable of “Technological Pedagogical Content Knowledge (TPACK)” level.

Technological Pedagogical Content Knowledge (TPACK) Level	<i>n</i>	so	sd	χ^2	<i>p</i>
Low	133	195.79			
Medium	273	251.67	2	153,042	0.001 *
High	159	409.75			
Total	565				

* Significant *p*-value at 0.05 level.

“Dunn’s post hoc test” with “Bonferroni” correction was conducted to identify the source of the significant difference found in the “Kruskal–Wallis H” test ($\chi^2(2) = 145,559$; $p = 0.001$). The results indicated that all three groups—low, medium, and high—differed significantly from one another ($p < 0.0166$). Based on the mean ranks, the high-TPACK-level group exhibited the highest scores, followed significantly by the medium- and low-level groups, respectively.

6. Findings Related to the Sub-Dimension of “Empowering Learners” of Music Teachers’ “Digital Competence” Levels

6.1. Findings Related to the Variable of Status of Possessing Internet and Technological Materials in the Sub-Dimension of “Empowering Learners”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “empowering learners” show a statistically significant difference according to the variable of status of possessing internet and technological materials, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

When Table 12 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “empowering learners” show a statistically significant difference in terms of the variable of status of possessing internet and technological materials ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “empowering learners,” music teachers who only have internet access have statistically higher “digital competences” compared to other music teachers.

Table 12. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “empowering learners” in terms of the variable of the status of possessing internet and technological materials.

Internet and Technological Material Ownership Status	<i>n</i>	so	sd	χ^2	<i>p</i>
Only Internet	34	308.66			
Only Technological Material	9	112.61	2	10,867	0.004 *
Both	522	284.27			
Total	565				

* Significant *p*-value at 0.05 level.

6.2. Findings Related to the Variable of “Technological Pedagogical Content Knowledge (TPACK)” Level in the Sub-Dimension of “Empowering Learners”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “empowering learners” show a statistically significant difference according to the variable of “TPACK” level, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

When Table 13 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “empowering learners” show a statistically significant

difference in terms of the variable of “TPACK” level ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “empowering learners,” music teachers with a high level of “TPACK” have statistically higher “digital competences” compared to other music teachers.

Table 13. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “empowering learners” in terms of the variable of “Technological Pedagogical Content Knowledge (TPACK)” level.

Technological Pedagogical Content Knowledge (TPACK) Level	<i>n</i>	so	sd	χ^2	<i>p</i>
Low	133	183.84			
Medium	273	267.87	2	124,381	0.001 *
High	159	391.92			
Total	565				

* Significant *p*-value at 0.05 level.

“Dunn’s post hoc test” with “Bonferroni” correction was conducted to determine the source of the significant difference identified by the “Kruskal–Wallis H” test ($\chi^2(2) = 148,016$; $p = 0.001$). The results indicated that all three groups—low, medium, and high—differed significantly from one another ($p < 0.0166$). Examination of the mean ranks revealed that professional engagement scores increased significantly as TPACK levels rose, with the high-level group exhibiting the highest scores.

7. Findings Related to the Sub-Dimension of “Facilitating Learners’ Digital Competence” of Music Teachers’ “Digital Competence” Levels

7.1. Findings Related to the Variable of Status of Possessing Internet and Technological Materials in the Sub-Dimension of “Facilitating Learners’ “Digital Competence”

In order to determine whether the “digital competence” levels of music teachers in the sub-dimension of “facilitating learners’ digital competence” show a statistically significant difference according to the variable of status of possessing internet and technological materials, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

When Table 14 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “facilitating learners’ digital competence” show a statistically significant difference in terms of the variable of status of possessing internet and technological materials ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “facilitating learners’ digital competence,” music teachers who possess both internet and technological materials have statistically higher “digital competences” compared to other music teachers.

Table 14. “Kruskal–Wallis H test” results of the findings obtained from the sub-dimension of “facilitating learners’ digital competence” in terms of the variable of status of possessing internet and technological materials.

Internet and Technological Material Ownership Status	<i>n</i>	so	sd	χ^2	<i>p</i>
Only Internet	34	242.26			
Only Technological Material	9	148.00	2	8848	0.012 *
Both	522	287.98			
Total	565				

* Significant *p*-value at 0.05 level.

7.2. Findings Related to the Variable of “Technological Pedagogical Content Knowledge (TPACK)” Level in the Sub-Dimension of “Facilitating Learners’ Digital Competence”

In order to determine whether the digital competence levels of music teachers in the sub-dimension of “facilitating learners’ digital competence” show a statistically significant difference according to the variable of “TPACK” level, the “Kruskal–Wallis H” test, one of the non-parametric tests, was applied.

When Table 15 is examined, it is seen that the “digital competence” levels of music teachers in the sub-dimension of “facilitating learners’ digital competence” show a statistically significant difference in terms of the variable of “TPACK” level ($p < 0.05$). As a result of examining the data in the table, it is observed that, according to the sub-dimension of “facilitating learners’ digital competence,” music teachers with a high level of “TPACK” have statistically higher “digital competences” compared to other music teachers.

Table 15. Kruskal–Wallis H test results of the findings obtained from the sub-dimension of “facilitating learners’ digital competence” in terms of the variable of “Technological Pedagogical Content Knowledge (TPACK)” level.

Technological Pedagogical Content Knowledge (TPACK) Level	<i>n</i>	so	sd	χ^2	<i>p</i>
Low	133	183.53			
Medium	273	270.17	2	118,381	0.001 *
High	159	388.24			
Total	565				

* Significant *p*-value at 0.05 level.

“Dunn’s post hoc test” with “Bonferroni” correction was conducted to identify the source of the significant difference found in the “Kruskal–Wallis H” test ($\chi^2(2) = 181,870$; $p = 0.001$). The results indicated that all three groups—low, medium, and high—differed significantly from one another ($p < 0.0166$). Examination of the mean ranks revealed that the scale total scores increased in parallel with TPACK levels, with each level having significantly higher scores than the preceding one.

8. Discussion

The findings of this study provide critical insights into the sustainability of digital transformation among music teachers in Türkiye, analyzed through the DigCompEdu framework and the TPACK model. The results indicate that music teachers primarily occupy the “explorer (B1)” level (mean = 33.65). This positioning aligns with international literature; for instance, Garzón-Artacho et al. [7] in Spain and Ivanov et al. [12] in Serbia both observed that while educators possess basic digital awareness, they remain at the “explorer” stage due to deficiencies in strategic integration and complex problem-solving. This suggests that music teachers recognize the potential of technology but encounter structural or pedagogical barriers in transforming this motivation into systematic classroom practices [31,33].

A pivotal finding of this research is the decisive role of the TPACK model as the primary driver of digital competence. Teachers with high TPACK scores demonstrated significantly superior digital proficiency across all sub-dimensions. This confirms the thesis by Mishra and Koehler [26] that technology integration is not merely about technical skill but occurs at the intersection of “content”, “pedagogy”, and “technology”. As Hattie [27] asserts, pedagogical competence is the strongest predictor of student success. Therefore, enhancing TPACK is essential for aligning music education with Sustainable Development Goal 4 (Quality Education).

Furthermore, this study highlights a significant shift in digital culture: the primacy of “access” over “ownership.” Interestingly, teachers with “only internet access” outperformed those with extensive hardware in several dimensions. This suggests that digital competence is more closely linked to “motivational agility” and active information seeking than to static physical assets. While hardware remains necessary for “facilitating” student competencies, internet-driven “lifelong learning” practices [7,12] appear to be more effective catalysts for teacher development.

Despite these drivers, a critical gap remains in “assessment” and “digital resource management.” The lack of significant difference in assessment scores—regardless of hardware ownership—indicates a universal training void. Music teachers struggle to integrate e-portfolios or real-time feedback tools into their curricula, a challenge also noted by Ivanov et al. [12]. This institutionalized gap necessitates a shift from technical training to “strategist” level professional development [31] to bridge the digital divide and ensure inclusive education [11].

In conclusion, these results demonstrate that while music teachers are receptive to the OECD [37] Education 2030 vision, they require targeted support to transition from “explorers” to “strategists.” Aligning teacher competencies with Target 4.7 of the 2030 Agenda requires a holistic redesign of curricula that blends technical resources with ethical digital citizenship [1,5]. Overcoming the technical infrastructure requirements for “facilitation” while addressing the pedagogical gaps in assessment will be the hallmark of a sustainable digital transformation in music education.

9. Conclusions and Recommendations

This study provides critical insights into the sustainability of digital transformation by examining the digital competencies of music teachers in Türkiye through the lenses of the “DigCompEdu” framework and the TPACK model. Empirical evidence indicates that the digital proficiency levels of music educators are predominantly clustered at the “explorer” (A2) stage. This finding suggests that while teachers recognize the instructional potential of digital technologies and have begun integrating basic tools into their curriculum, they tend to utilize these technologies to supplement traditional methods rather than fundamentally restructuring pedagogical processes [2,3].

A primary contribution of this research is the emergence of TPACK as the most consistent and distinctive predictor across all dimensions of digital competence [3]. The significantly higher digital proficiency observed in teachers with advanced TPACK levels corroborates the theoretical premise that educational success with technology depends not on isolated technical skills (TK), but on the depth of their integration with musical content knowledge (CK) and pedagogy (PK) [4]. Consequently, the TPACK model serves as a foundational structural element in developing digital competencies, necessitating the translation of existing technological awareness into sustainable professional development.

Another striking finding of the study is the “internet access and digital competence paradox.” The data reveal that, across several dimensions, teachers with only internet access demonstrated higher proficiency than those possessing both internet and comprehensive physical hardware [5,6]. This aligns with contemporary critiques of “techno-solutionism,” suggesting that an abundance of hardware does not inherently yield pedagogical excellence; rather, the significant difference arises from self-directed learning and information-seeking strategies facilitated by the internet [1,6]. Conversely, the superiority of teachers with both internet and hardware in the “developing students’ digital competencies” dimension confirms that physical infrastructure remains a vital supplementary condition for students’ digital content creation processes.

Based on these findings, the following recommendations are proposed for a sustainable digital transformation in music education:

- **Pedagogy-Oriented Training:** In-service training should move beyond software instruction to an integrated structure focusing on the synthesis of technology with musical content knowledge (CK) and pedagogy (PK), as defined by TPACK.
- **Digital Mentorship Systems:** Professional Learning Communities (PLCs) should be established to facilitate sharing networks where “integrator” (B1) and “expert” (B2) level teachers provide guidance to “explorer” (A2) colleagues.
- **Encouraging Assessment Software:** The use of digital assessment tools (e-portfolios, learning analytics, etc.) should be incentivized within the curriculum to overcome stagnation in this dimension and enhance teachers’ ability to use data for performance improvement.

As highlighted by John Hattie [27], a teacher’s pedagogical competence is the most potent factor influencing student achievement. Digital transformation in music education is not merely a matter of equipping classrooms with hardware; it is about the teacher’s capacity to manage these tools with artistic creativity and pedagogical depth. Moving beyond being technical users (explorers) to becoming leaders and pioneers who design digital processes—consistent with UNESCO [17] lifelong learning principles—is an imperative for a sustainable future. Supported by international standards and theoretical models, the potential of music teachers to adapt to the digital age remains the strongest guarantee for a high-quality and inclusive future music education system.

10. Limitations and Suggestions for Future Research

As with any scientific inquiry, this study should be evaluated within the context of certain limitations. The primary constraints and their implications for future academic endeavors are outlined below.

- **Methodological Framework:** This study utilizes a cross-sectional design with data collected through a quantitative approach and self-report scales. Discrepancies may exist between teachers’ perceived competencies and their actual classroom practices. Future research could utilize mixed-methods designs—incorporating observations, interviews, or portfolio analysis—to capture the practical application of digital competencies more profoundly.
- **Sample Scope:** The study is limited to music teachers within Türkiye. Given that digital transformation is influenced by cultural and economic variables, conducting cross-cultural comparative studies would provide a broader perspective on the universal validity of the DigCompEdu and TPACK models.
- **Need for Longitudinal Analysis:** The current study provides a snapshot of digital competence at a specific point in time. In an era of rapidly evolving technological tools and educational policies, longitudinal research tracking the development of digital proficiency over time would more clearly elucidate the sustainability of these skills and their correlation with professional seniority.
- **Necessity of Experimental Studies:** While research results descriptively prove the impact of TPACK levels on digital competence, future studies should implement quasi-experimental designs to test the effectiveness of TPACK-based in-service training programs on teachers’ digital skills, thereby filling a gap in the literature regarding application.
- **Correlation with Student Outcomes:** This research is teacher-centered. Future studies investigating the direct relationship between teachers’ digital proficiency levels and students’ musical achievement, motivation, or digital literacy would be a critical step in measuring the educational efficacy of digital transformation.

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